

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

VECTOR

LECTURE-2

PHYSICS

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Topics To Be Covered

- 1 Revision ✓
- 2 ~~Law of~~ Parallelogram Law of Vector Addition ✓
- 3 Multiplication of Two vector ✓
- 4 ..
- 5 ..

“You get what you give”

Recap from 1st Lecture

Vector

- The $\vec{p}$ have magnitude and direction



A

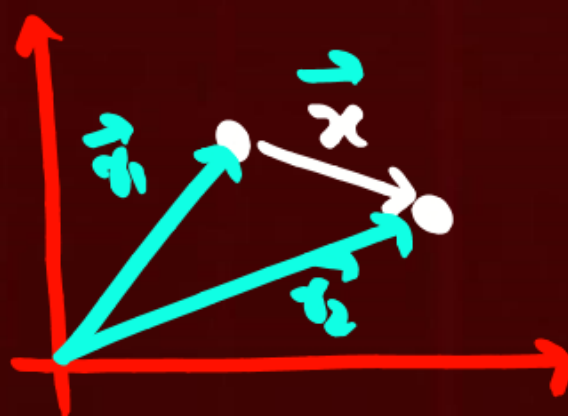
$\vec{A}$

Types of Vector

(i) Position Vector



(ii) Displacement V



(iii) Parallel Vector

$$\vec{A} \parallel \vec{B} \quad \theta = 0$$

(iii) Antiparallel Vector



(iv) Equal Vector

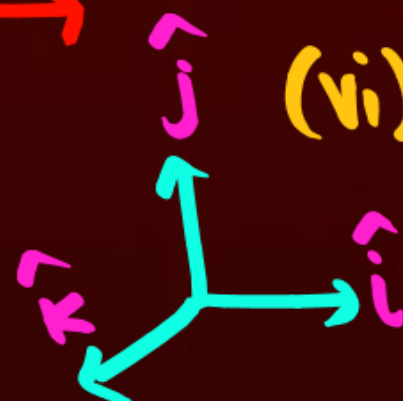
$$\vec{A} = \vec{A}$$

(v) Orthogonal Vector



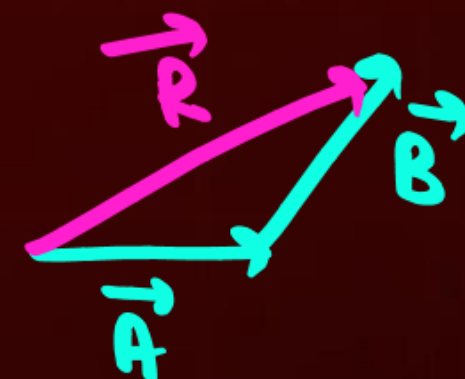
(vi) Unit Vector

$n=1$
Direction



+ A
- B
x C
÷ D
a
vector

① Triangular law of Vector Addition

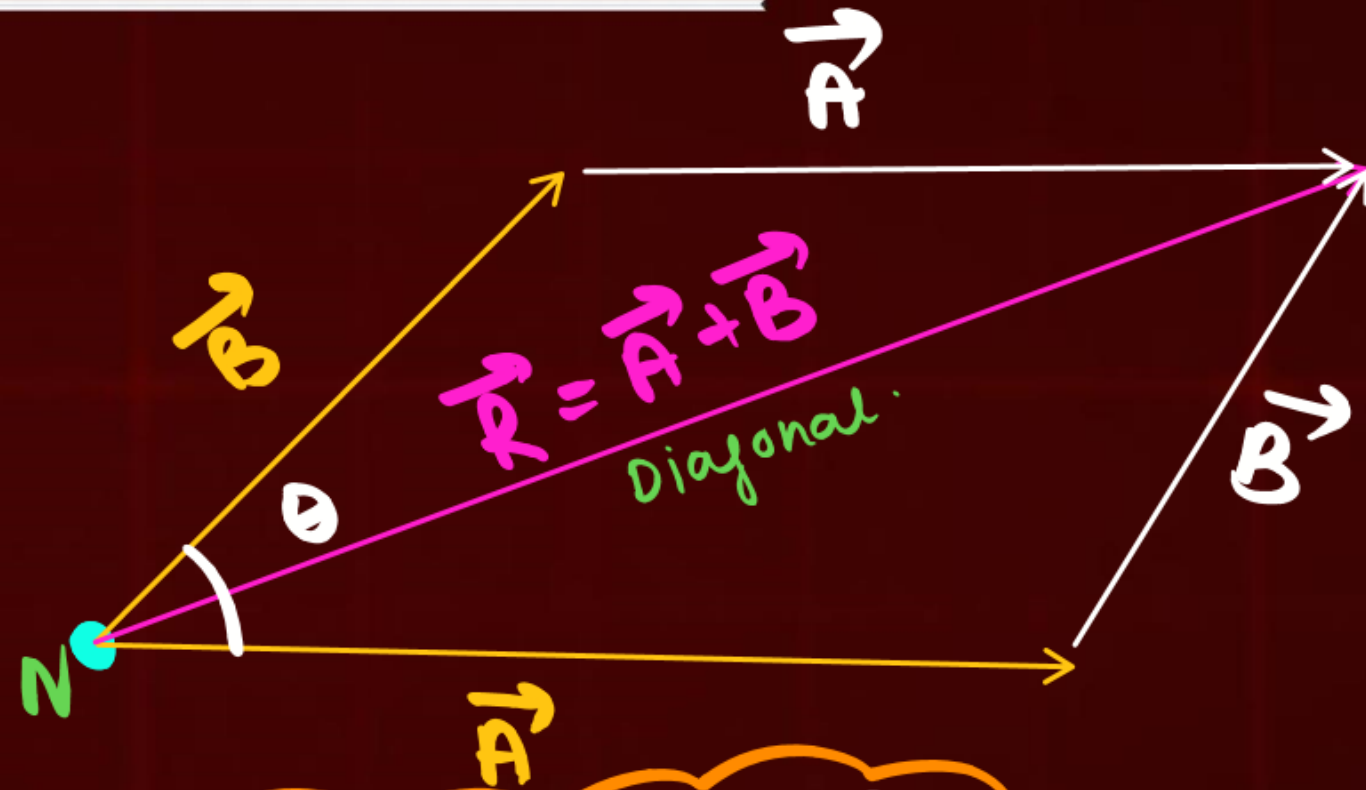


Law of Vector Addition

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

2. Parallelogram law of vector addition

If two vectors are the Adjacent Side of llgm the Diagonal is the Resultant Vector



Note:- vector ko kahi B laa jaya jasakta hai without changing the Direction.

Boared

2025, 2024

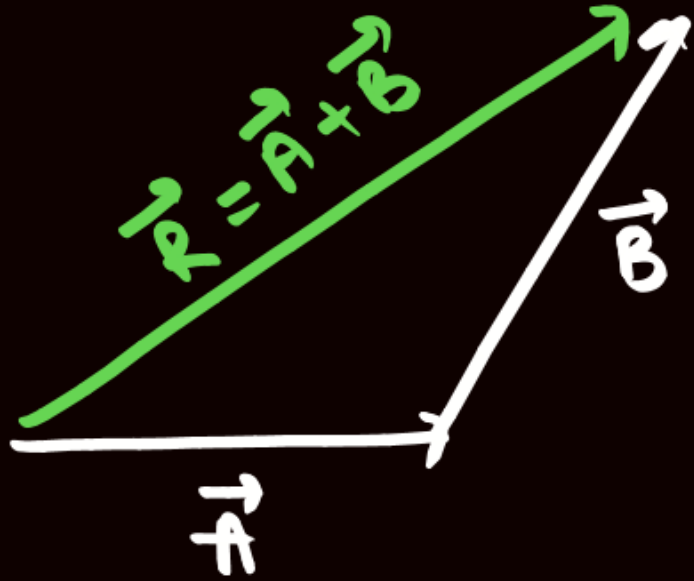
2023

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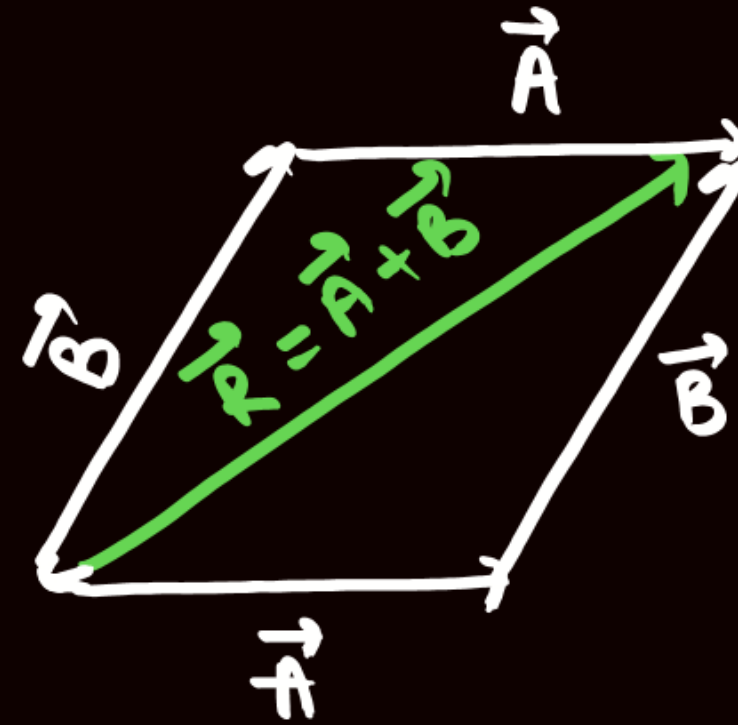
2022



(i) Triangular law of Vector Addition

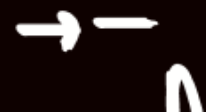


(ii) llgm law of vector addition

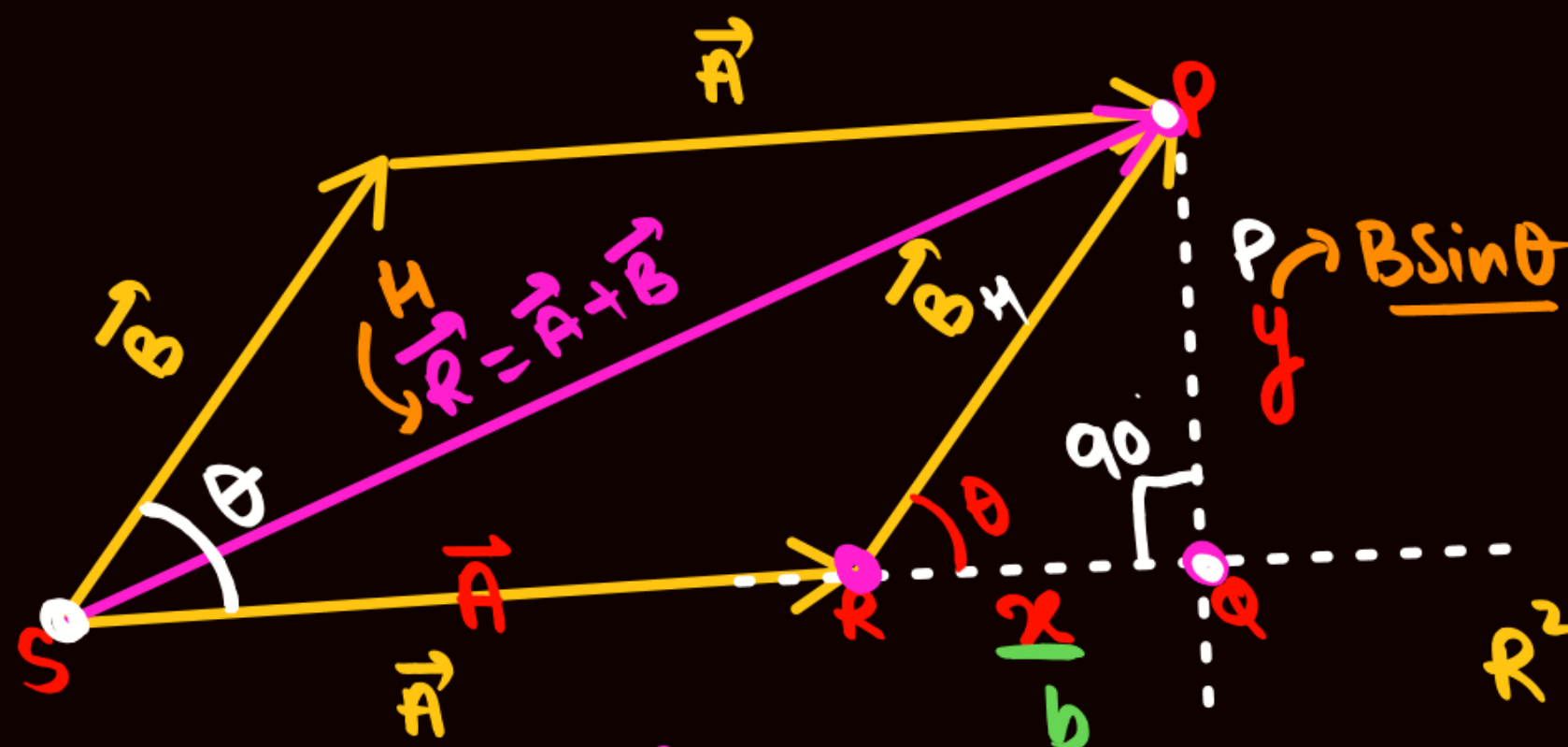


$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

tan α



Resultant vector $\vec{R}$ $(A+B)^2 = A^2 + B^2 + 2AB \cos \theta$



In small ΔPOR

$\sin \theta = \frac{p}{H}$

$\sin \theta = \frac{y}{B}$

$B \sin \theta = y$

$\cos \theta = \frac{B}{H}$

$\cos \theta = \frac{x}{B}$

$B \cos \theta = x$

In big ΔPQS (By PQT)

$H^2 = B^2 + p^2$

$R^2 = (A+x)^2 + y^2$

$R^2 = (A+B \cos \theta)^2 + (B \sin \theta)^2$

$R^2 = A^2 + (B \cos \theta)^2 + 2AB \cos \theta + B^2 \sin^2 \theta$

$R^2 = A^2 + B^2 \sin^2 \theta + B^2 \cos^2 \theta + 2AB \cos \theta$

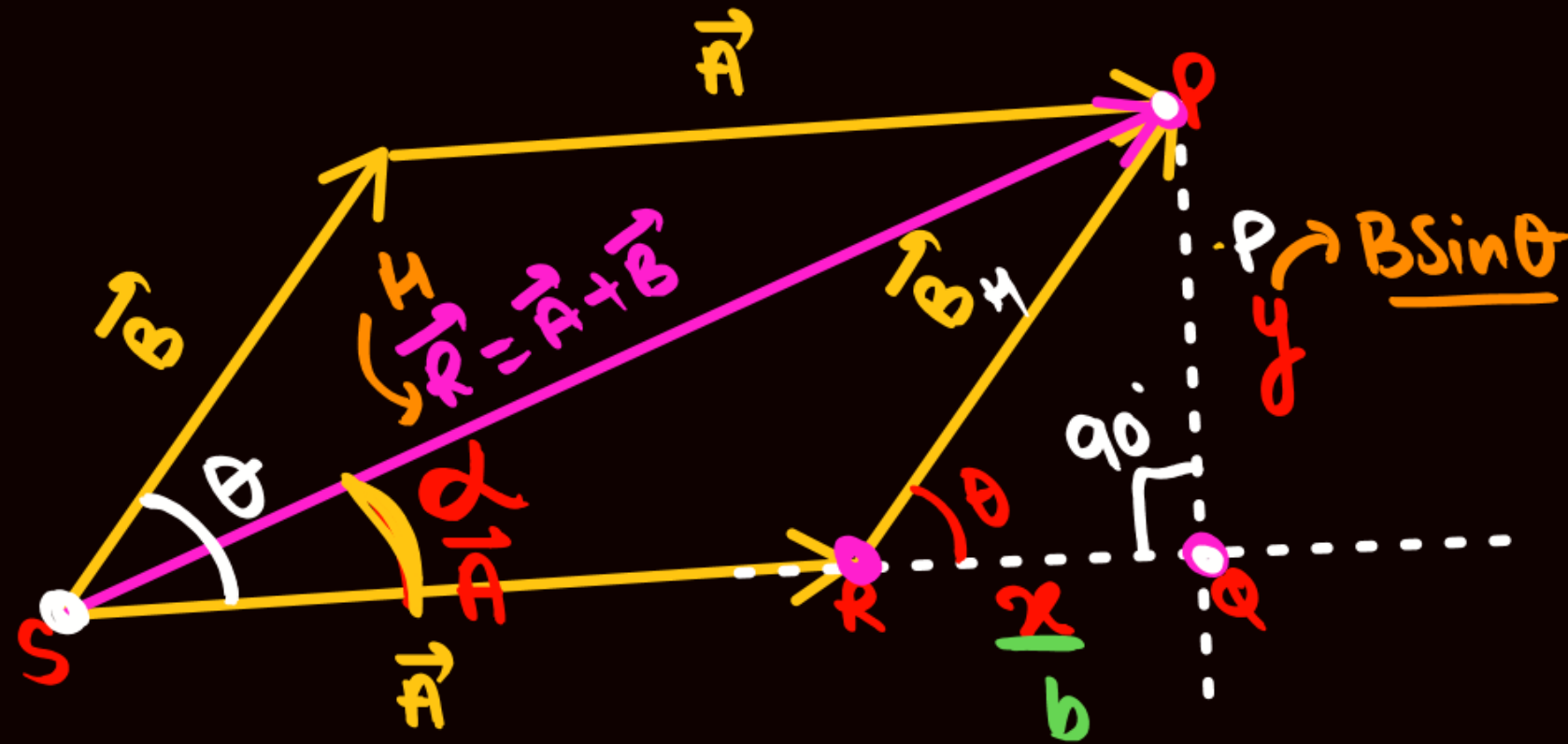
$R^2 = A^2 + B^2 (\sin^2 \theta + \cos^2 \theta) + 2AB \cos \theta$

$R^2 = A^2 + B^2 \times 1 + 2AB \cos \theta$

$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Resultant vector $\vec{R}$ $(A+B)^2 = A^2 + B^2 + 2AB \cos \theta$



In $\triangle PQR$

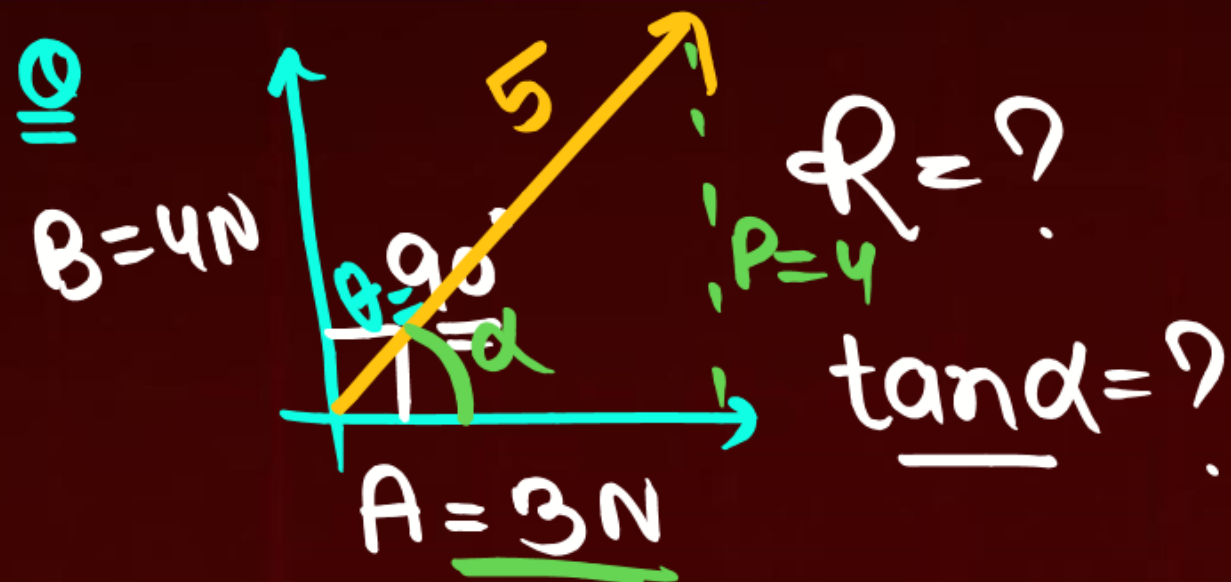
$$\tan \alpha = \frac{P}{B} = \frac{y}{A+x}$$

$$\tan \alpha = \frac{y}{A+x}$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

#

Numericals



(a) ~~5N~~

(b) 3N

(c) 2N

(d) 1N

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta} \rightarrow 90^\circ$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
3 4 3 4

$$R = \sqrt{(3)^2 + (4)^2 + 2 \times 3 \times 4 \cos 90^\circ}$$

$$R = \sqrt{9 + 16 + 0}$$

$$R = \sqrt{25}$$

$$R = \sqrt{5^2} = 5$$

$$\tan \alpha = \frac{P}{B} = \frac{4}{3}$$

SUMMARY

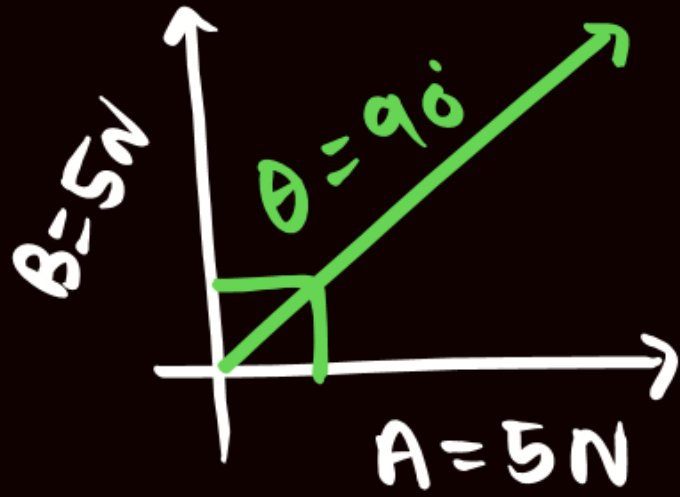


llgm law of vector addition

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

~~4/10~~ Numerical - (2)



$$R = ?$$

$$\tan \theta = ?$$

$$R_s: \underline{10}$$

Thank
You